

Heaven's Light is Our Guide



Rajshahi University of Engineering and Technology

Dept. of Electrical and Computer Engineering

Course Code: ECE-2112

Course Title: Digital Techniques Sessional

Lab Report 02

Simplification of Boolean Expressions and Verification

Using Logic Circuit Simulation

Date of Submission: 16 August, 2026

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Session: 2024-2025

Table of Contents

Table of Contents	2
Task 01: Simplification and Verification of Boolean Expression $F(A, B, C) = A'BC + AB'C + ABC' + ABC$...	3
Objective.....	3
Algebraic Simplification	3
Truth Table	3
Karnaugh Map (K-Map).....	3
Circuit Diagram	4
Discussion.....	4
Conclusion.....	4
Task 02: Simplification and Verification of the Boolean Expression $F(A, B, C) = A(A+B)(A+B+C)$	5
Objective.....	5
Algebraic Simplification	5
Truth Table	5
Karnaugh Map	5
Circuit Diagram	6
Discussion.....	6
Conclusion	6
Task 03: Simplification and Verification of the Boolean Expression $F(A, B, C) = (A + (BC)')'(AB + ABC)$	7
Objective.....	7
Algebraic Simplification	7
Truth Table	7
Karnaugh Map	7
Circuit Diagram	8
Discussion.....	8
Conclusion	8
Task 04: Simplification and Verification of the Boolean Expression $F(A, B) = (B'(A + B) + (A + B)(A + B'))B'$	9
Objective.....	9
Algebraic Simplification	9
Truth Table	9
Karnaugh Map	9
Circuit Diagram	10
Discussion.....	10
Conclusion.....	10
References	10

Task 01:

Simplification and Verification of Boolean Expression $F(A, B, C) = A'BC + AB'C + ABC' + ABC$

Objective

To reduce a three-variable sum-of-minterms expression to its minimal sum-of-products form, to confirm the reduction using a Karnaugh map, and to verify that the original and simplified circuits are functionally identical by implementing both in Logisim.

Algebraic Simplification

$$\begin{aligned} F(A, B, C) &= A'BC + AB'C + ABC' + ABC \\ &= A'BC + ABC + AB'C + ABC + ABC' + ABC \\ &= BC(A' + A) + AC(B' + B) + AB(C' + C) \\ &= BC + AC + AB = AB + BC + AC \end{aligned}$$

Truth Table

A	B	C	F = AB + BC + AC
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Table 01: Truth table for $F(A, B, C) = AB + BC + AC$

Karnaugh Map (K-Map)

Three groups of two adjacent 1-cells are formed: the pair at $BC = 11$ (both rows) gives BC ; the pair at $A = 1, C = 1$ ($BC = 01, 11$) gives AC ; and the pair at $A = 1, B = 1$ ($BC = 11, 10$) gives AB . Combining the three groups yields the minimized expression $F = AB + AC + BC$, matching the algebraic result.

Circuit Diagram

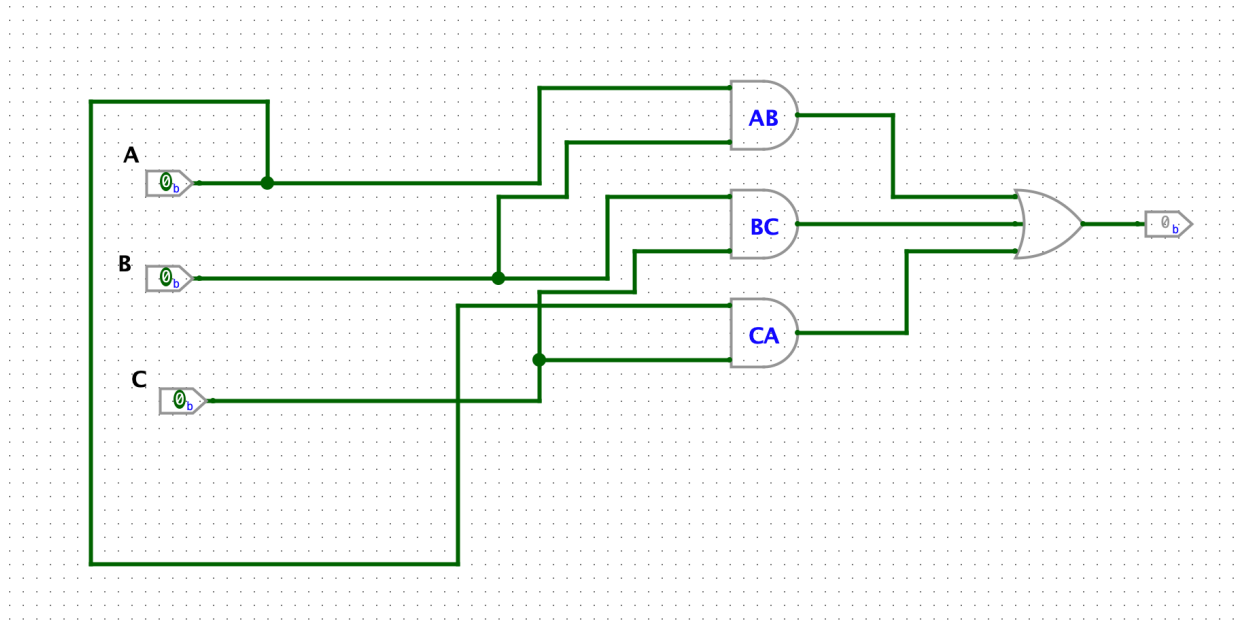


Figure 01: Circuit Diagram for $F(A, B, C) = A'BC + AB'C + ABC' + ABC$ (original circuit on top, simplified circuit below)

Discussion

The original circuit uses four 3-input AND gates, inverters, and an OR gate. After simplification, the circuit becomes $AB + AC + BC$, requiring only three 2-input AND gates and an OR gate. Logisim simulation showed that both circuits produced identical outputs, confirming the simplification.

Conclusion

The expression $F = A'BC + AB'C + ABC' + ABC$ was simplified to $F = AB + AC + BC$ using Boolean algebra and a Karnaugh map. Simulation verified that the simplified circuit gives the same output as the original for all input combinations.

Task 02:

Simplification and Verification of the Boolean Expression $F(A, B, C) = A(A+B)(A+B+C)$

Objective

To reduce a three-variable product-of-sums-type expression to its simplest form, to confirm the reduction using a Karnaugh map, and to verify the equivalence of the original and simplified circuits in Logisim.

Algebraic Simplification

$$F(A, B, C) = A(A + B)(A + B + C)$$

$$= (A + AB)(A + B + C)$$

$$= A(1 + B)(A + B + C)$$

$$= A(A + B + C) = A + AB + AC$$

$$= A(1 + B + C) = A$$

Truth Table

A	B	C	F = A
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Table 03: Truth table for $F(A, B, C) = A$

Karnaugh Map

All four cells in the $A = 1$ row are 1, while the entire $A = 0$ row is 0. This forms a single group of four adjacent cells covering the whole row, which eliminates both B and C and leaves only $F = A$, exactly matching the algebraic simplification.

Circuit Diagram

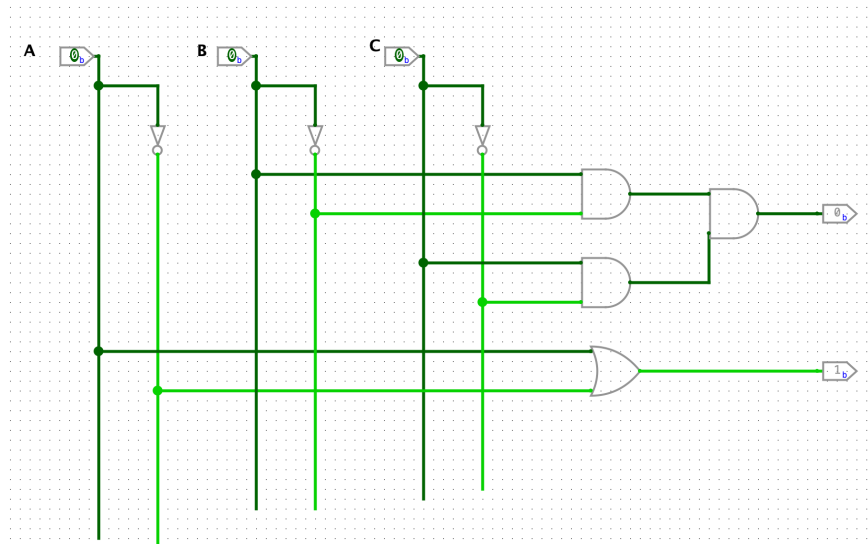


Figure 02: Circuit Diagram for $F(A, B, C) = A(A+B)(A+B+C)$ (original circuit on top, simplified circuit — a direct wire — below)

Discussion

The original circuit implements $F = A(A + B)(A + B + C)$ using OR and AND gates. Applying the absorption law simplifies the expression to $F = A$, eliminating all logic gates. Logisim simulation confirmed that the output always equals A, regardless of B and C.

Conclusion

The expression $F = A(A + B)(A + B + C)$ was simplified to $F = A$ using the absorption law. Karnaugh map analysis and Logisim simulation verified that the simplified circuit produces the same output as the original for all input combinations.

Task 03:

Simplification and Verification of the Boolean Expression $F(A, B, C) = (A + (BC)')'(AB + ABC)$

Objective

To reduce a mixed AND-OR-complement expression using De Morgan's theorem and Boolean algebra, to confirm the result with a Karnaugh map, and to verify using Logisim that the simplified circuit behaves as a constant.

Algebraic Simplification

$$F(A, B, C) = (A + (BC)')'(AB + ABC)$$

$$= A' \cdot BC \cdot (AB + ABC)$$

$$= A'BC \cdot AB(1 + C) = A'BC \cdot AB$$

$$= (A \cdot A') \cdot BC = 0$$

Truth Table

A	B	C	F = 0
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

Table 05: Truth table for $F(A, B, C) = 0$

Karnaugh Map

Every cell of the map is 0, so there are no 1-cells to group. A Karnaugh map with no populated cells represents the constant function $F = 0$, confirming the algebraic result.

Circuit Diagram

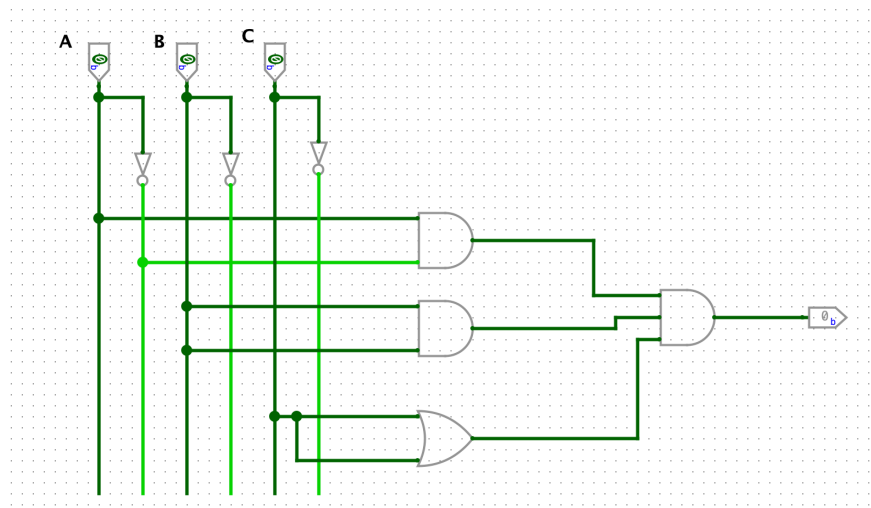


Figure 03: Circuit Diagram for $F(A, B, C) = (A + (BC)')'(AB + ABC)$ (original circuit on top, simplified circuit — constant 0 — below)

Discussion

Using De Morgan's theorem and the absorption law, the expression simplifies to 0 because it contains the term $A \cdot A' = 0$. Logisim simulation confirmed that the output remains 0 for all input combinations of A, B, and C.

Conclusion

The expression $F = (A + (BC)')'(AB + ABC)$ simplifies to 0. Boolean algebra, Karnaugh map analysis, and Logisim simulation all verified that the output is always logic 0.

Task 04:

Simplification and Verification of the Boolean Expression $F(A, B) = (B'(A + B) + (A + B)(A + B'))B'$

Objective

To reduce a two-variable expression built from nested OR and AND terms to its minimal form, to confirm the reduction using a Karnaugh map, and to verify the equivalence of the original and simplified circuits in Logisim.

Algebraic Simplification

$$\begin{aligned} F(A, B) &= (B'(A + B) + (A + B)(A + B'))B' \\ &= B'(AB' + BB' + AA + AB' + AB + BB') \\ &= B'(AB' + A + AB' + AB) \\ &= AB'(B' + 1 + B' + B) = AB' \end{aligned}$$

Truth Table

A	B	$F(A, B) = AB'$
0	0	0
0	1	0
1	0	1
1	1	0

Table 07: Truth table for $F(A, B) = AB'$

Karnaugh Map

Only a single cell, at $A = 1, B = 0$, is populated. A lone 1-cell cannot be combined with any adjacent cell, so it is read directly as the product of its coordinate literals, giving $F = AB'$, matching the algebraic result.

Circuit Diagram

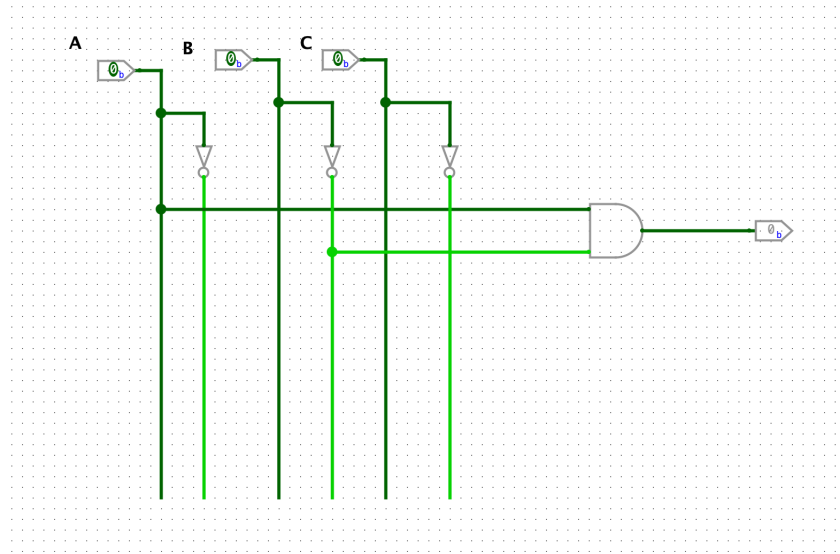


Figure 04: Circuit Diagram for $F(A, B) = (B'(A+B) + (A+B)(A+B'))B'$ (original circuit on top, simplified circuit — a single AND gate with B inverted — below)

Discussion

Using Boolean algebra, the expression simplifies to $F = AB'$. The simplified circuit requires only an inverter on B and a 2-input AND gate. Logisim simulation confirmed that both the original and simplified circuits produce identical outputs for all input combinations.

Conclusion

The expression $F = (B'(A + B) + (A + B)(A + B'))B'$ was simplified to $F = AB'$. The result was verified using Boolean algebra, a Karnaugh map, and Logisim simulation.

References

- [1] M. M. Mano and M. D. Ciletti, Digital Design, 5th ed. Upper Saddle River, NJ, USA: Pearson, 2013.
- [2] R. J. Tocci, N. S. Widmer, and G. L. Moss, Digital Systems: Principles and Applications, 11th ed. Upper Saddle River, NJ, USA: Pearson, 2011.